

Course Level: Undergraduate Open Elective (B. Tech)

BL: 2 Understand

CO: Recall the basics of food groups, food composition, nutritive value and functions of food.

Module 2: Micronutrients: Vitamins

1. Introduction

Vitamins = micronutrients, low molecular weight biomolecules.

Required for growth, development, and metabolism.

13 essential vitamins: some water-soluble, some fat-soluble.

Body synthesizes few; most must come from diet.

Fruits, vegetables, whole grains, and healthy fats provide most vitamins.

2. Fat-Soluble Vitamins

Stored in fatty tissues & liver; last days to months.

Vitamin A (Retinol)

Functions: vision, antioxidant, bone growth, immune system, cancer risk reduction.

Sources: liver, cod liver oil, carrots, spinach, sweet potatoes, eggs, milk.

Deficiency: night blindness, dry eyes/skin.

Vitamin D (Calciferol)

Functions: calcium & phosphorus absorption, bone health, immune regulation.

Sources: sunlight (D3), cod liver oil, salmon, egg yolk, fortified milk.

Deficiency: rickets, osteomalacia

Vitamin E (Tocopherol)

Functions: antioxidant, immune support, prevents arterial clots.

Sources: wheat germ oil, sunflower seeds, almonds, spinach, mango.

Deficiency: Infertility

Vitamin K (Phylloquinone, Menaquinone)

Functions: blood clotting (prothrombin), bone health.

Sources: leafy greens, soybean oil, fermented foods, meat, eggs.

Deficiency: bleeding

3. Water-Soluble Vitamins

Not stored; need regular intake.

Vitamin C (Ascorbic acid)

Functions: collagen synthesis, wound healing, antioxidant.

Sources: citrus fruits, strawberries, broccoli, bell peppers.

Deficiency: scurvy, gum bleeding, delayed wound healing.

Vitamin B Complex

B1 (Thiamine): energy metabolism, nerve function. Sources: pork, grains, eggs. Deficiency: weight loss, neuropathy.

B2 (Riboflavin): growth, metabolism. Sources: milk, eggs, green beans. Deficiency: cracked lips, anemia.

B3 (Niacin): cell growth, nervous system. Sources: chicken, tuna, nuts. Deficiency: depression, fatigue.

B5 (Pantothenic acid): hormone & energy production. Sources: meat, mushrooms, yogurt. Deficiency: cramps, irritability.

B6 (Pyridoxine): neurotransmitter synthesis, RBC formation. Sources: chickpeas, bananas, beef liver. Deficiency: anemia, depression.

B7 (Biotin): metabolism, hair/skin health. Sources: yeast, eggs, leafy veg. Deficiency: brittle nails, rashes.

B9 (Folic acid): DNA/RNA synthesis, RBC production. Sources: leafy greens, peanuts, liver. Deficiency: anemia, irregular heartbeat.

B12 (Cobalamin): brain function, RBC formation, DNA regulation. Sources: dairy, meat, fish, eggs. Deficiency: pernicious anemia, fatigue, memory loss.

4. Vitamin Deficiency Overview

Deficiency leads to varied disorders: anemia, neuropathy, scurvy, osteoporosis, impaired immunity.

Vitamin B17 (Amygdalin) – controversial, plant-derived compound (beans, seeds, legumes); claimed pain-relief & immunity benefits but not universally accepted.

#Vitamins_Deficiency

- 1- Vitamin "A": Night blindness
- 2- Vitamin "B1": Beriberi
- 3- Vitamin "B2": Ariboflavinosis
- 4- Vitamin "B3": Pellagra
- 5- Vitamin "B5": Parestheia
- 6- Vitamin "B6": Anemia
- 7- Vitamin "B7": Dermatitis and enteritis
- 8- Vitamin "B9" - "B12": Megaloblastic anerr
- 9- Vitamin "B17": Cancer
- 10- Vitamin "C": Scurvy and swelling of gum
- 11- Vitamin "D": Rickets and Osteomalacia
- 12- Vitamin "E": less fertility
- 13- Vitamin "K": Non-Clotting of blood